Clarusway

Backend Workshop -1-

Workshop

Subject:

- Introduction to Backend Development

Learning Goals

- Having general information about the backend.

Introduction

- As developers, we should also be able to express what we know. This study was prepared to support this purpose.

Prerequisites

- We will use the VSCode you are familiar with.
- At the same time, we need to install Nodejs on our computer.

Lets start

1. What Is Backend Development ?

Answer:

2. What is skills a backend developer must have ?

Answer:

3. Why backend is important ?

Answer:

4. What is SQL?

Answer:

5. What is JOIN and what are its types?

Answer:

6. What is the difference between WHERE and HAVING clauses in SQL?

Answer:

7. What are the GROUP BY and HAVING clauses in SQL?

Answer:

8.What is the difference between PRIMARY KEY and FOREIGN KEY?

Answer:

9.What is SQL Injection and how can it be prevented?

Answer:

10.What is an INDEX in SQL and why is it used?

Answer:

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Teamwork

Subject: SQL

Learning Goals

- To be able to write SQL statements that will perform the desired query.

Introduction

We use the SQL language when performing operations on relational databases. You can perform many different operations on the DB with SQL, but this work only includes querying.

Lets start

Write SQL statements that produce the desired output.

1. WRITE A QUERY THAT RETURNS TRACK NAME AND ITS COMPOSER FROM TRACKS TABLE

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2. WRITE A QUERY THAT RETURNS ALL COLUMNS FROM TRACKS TABLE

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3. WRITE A QUERY THAT RETURNS THE UNIQUE NAME OF COMPOSERS OF EACH TRACK

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4. WRITE A QUERY THAT RETURNS UNIQUE ALBUMID, MEDIATYPEID FROM TRACKS TABLE

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5. WRITE A QUERY THAT RETURNS TRACK NAME AND TRACKID OF ‘Jorge Ben’

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6. WRITE A QUERY THAT RETURNS ALL INFO OF THE INVOICES OF WHICH TOTAL AMOUNT IS GREATER THAN $25

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7. WRITE A QUERY THAT RETURNS ALL INFO OF THE INVOICES OF WHICH TOTAL AMOUNT IS LESS THAN $15. JUST RETURN 5 ROWS

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8. WRITE A QUERY THAT RETURNS ALL INFO OF THE INVOICES OF WHICH TOTAL AMOUNT IS GREATER THAN $10. THEN SORT THE TOTAL AMOUNTS IN DESCENDING ORDER, LASTLY DISPLAY TOP 2 ROWS

9. WRITE A QUERY THAT RETURNS ALL INFO OF THE INVOICES OF WHICH BILLING COUNTRY IS NOT CANADA. THEN SORT THE TOTAL AMOUNTS IN ASCENDING ORDER, LASTLY DISPLAY TOP 10 ROWS

10. WRITE A QUERY THAT RETURNS INVOICEID, CUSTOMERID AND TOTAL DOLLAR AMOUNT FOR EACH INVOICE. THEN SORT THEM FIRST BY CUSTOMERID IN ASCENDING, THEN TOTAL DOLLAR AMOUNT IN DESCENDING ORDER.

11. WRITE A QUERY THAT RETURNS ALL TRACK NAMES THAT START WITH ‘B’ AND END WITH ‘S’

12. WRITE A QUERY THAT RETURNS THE NEWEST DATE AMONG THE INVOICE DATES BETWEEN 2008 AND 2011

13. WRITE A QUERY THAT RETURNS THE FIRST AND LAST NAME OF THE CUSTOMERS WHO HAVE ORDERS FROM NORWAY AND BELGIUM

14. WRITE A QUERY THAT RETURNS THE TRACK NAMES OF ‘ZAPPA’

15. HOW MANY TRACKS AND INVOICES ARE THERE IN THE DIGITAL MUSIC

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STORE, DISPLAY SEPERATELY
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16. HOW MANY COMPOSERS ARE THERE IN THE DIGITAL MUSIC STORE

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17. HOW MANY TRACKS DOES EACH ALBUM HAVE, DISPLAY ALBUMID AND NUMBER OF TRACKS SORTED FROM HIGHEST TO LOWEST

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18. WRITE A QUERY THAT RETURNS TRACK NAME HAVING THE MINIMUM AND MAXIMUM DURATION, DISPLAY SEPERATELY

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19. WRITE A QUERY THAT RETURNS THE TRACKS HAVING DURATION LESS THAN THE AVERAGE DURATION

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20. WRITE A QUERY THAT RETURNS THE TOTAL NUMBER OF EACH COMPOSER’s TRACK.

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21. WRITE A QUERY THAT RETURNS THE GENRE OF EACH TRACK.

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22. WRITE A QUERY THAT RETURNS THE ARTIST’s ALBUM INFO.

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23. WRITE A QUERY THAT RETURNS THE MINIMUM DURATION OF THE TRACK IN EACH ALBUM. DISPLAY ALBUMID, ALBUM TITLE AND DURATION OF THE TRACK. THEN SORT THEM FROM HIGHEST TO

LOWEST

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24. WRITE A QUERY THAT RETURNS ALBUMS WHOSE TOTAL DURATION IS HIGHER THAN 60 MIN. DISPLAY ALBUM TITLE AND THEIR DURATIONS. THEN SORT THE RESULT FROM HIGHEST TO LOWEST

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25. WRITE A QUERY THAT RETURNS TRACKID, TRACK NAME AND ALBUMID INFO OF THE ALBUM WHOSE TITLE ARE ‘Prenda Minha’, 'Heart of the Night' AND 'Out Of Exile'.

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☺ **Thanks for Attending** ✍

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